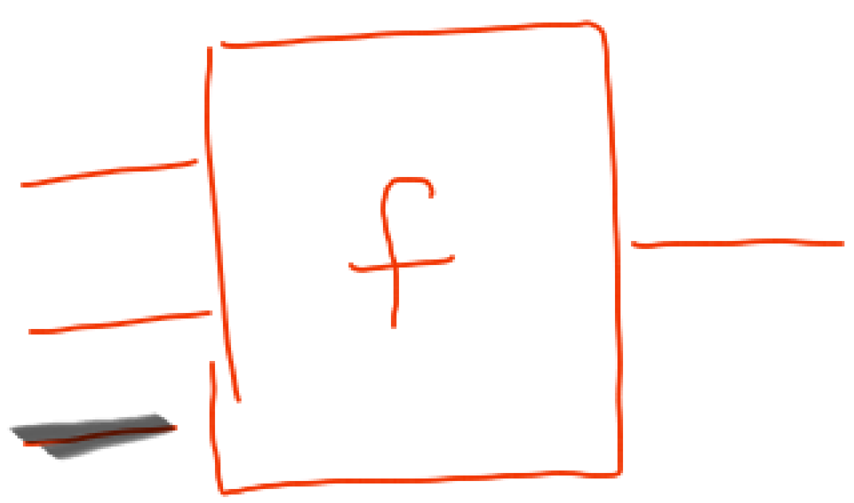
22.9.25

"Computer"

$N = \{1, 2, \ldots\}$

Inputs

Output



function

$N \to N$

$N^k \to N$

Language $\subseteq \Sigma^*$

$\{0,1\}$

Reg, CFG } Pumping Lemma

there is a language that a TM cannot ACCEPT

1 define

I.D.

TM

$$\overset{\text{tape}}{\alpha}\; q_k \;\overset{\text{right}}{\beta}$$

$$M = \langle Q, \Sigma, \Gamma, \delta, q_1, b\!\!\!/, F \rangle$$

δ: transitions

$$Q \times \Gamma \to Q \times \Gamma \times D$$

tape symbols

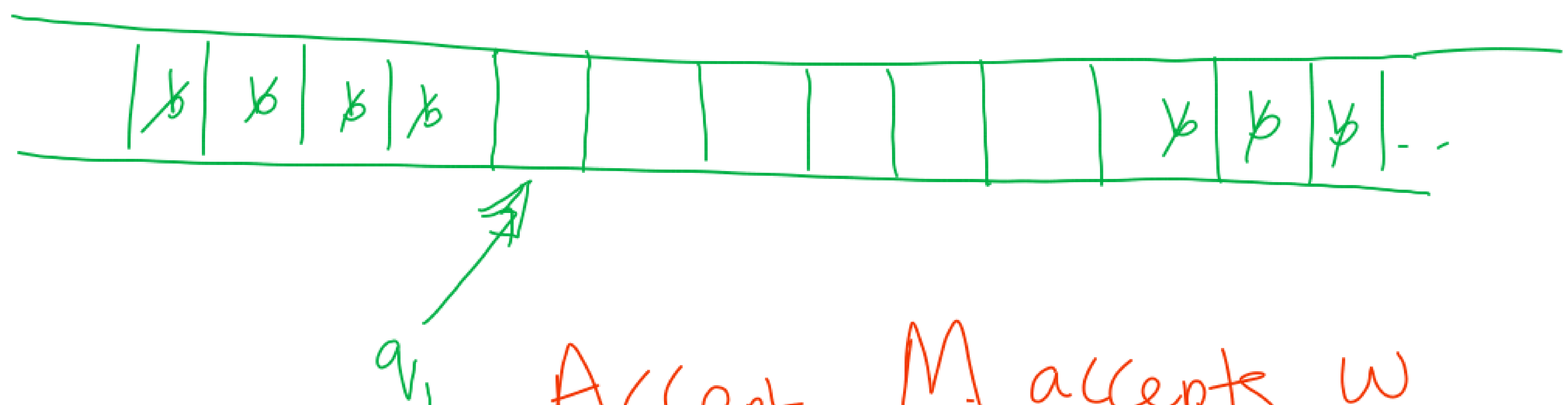
direction L, R

F: ◎

$q_1 w \vdash \cdots \vdash \;\cdots \vdash$ ◯

Start

w



State transitions

Accept

M accepts w

① Final state

② machine Halts

$F = \phi$ (no ◎ states)

$$q_0 w \vdash \cdots \vdash \alpha q_k \beta$$

$q_k \in F$ ◎ q_k

$$q_1 w \vdash \vdash \vdash \alpha q_k \beta \not\vdash$$

no further move

TM → Unique BitString

Encode a TM as a bitstring

$\Sigma = \{0, 1\}$

$a_1 \; a_2 \cdots a_k$

$\Gamma = \{X, Y \ldots\}$

$t_1 \; t_2 \ldots$

|||

$\| \sim\sim\sim \| \sim\sim\sim \| \; 0001\,0000000\,1\,000\,1\,00000\,1\,0 \; \|$

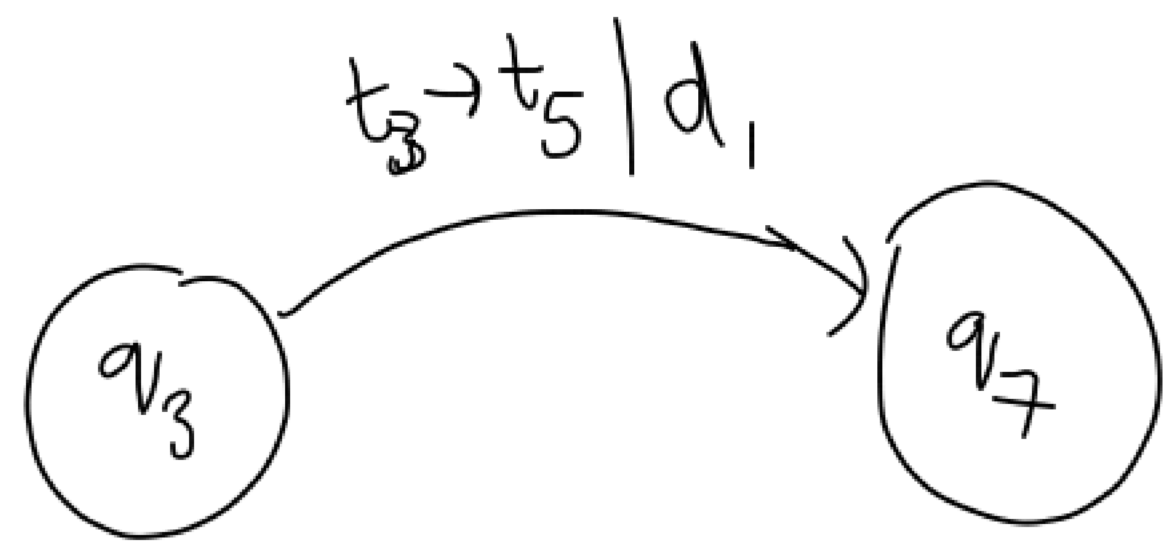
3 7 3 5 1

|||

simple move

$D = \{L, R\}$

$d_1 \quad d_2$



$\{1, 2, \ldots\}$

Bitstring $\longleftrightarrow$ $\mathbb{N}$ $\longleftrightarrow$ $\mathbb{N}^k$

Unique prime factorization

ϵ — 1 →

0 — 10 → 2

1 — 11 → 3

00 → 100 → 4

01 → 101 → 5

⋮

10 → 2

9 → 1001 → 9

01001

11001

101001

41 → 101001 → 41

11001

Binary form of Decimal

1	— ϵ		1
2	— 0		2
3	— 1		3
4	— 00		4
5	— 01		5
6	— 10		6
7	— 11		
8	— 000		
9	— 001		
10	— 010		
11	— 011		
	100		

leading 1

Decide $w \in L$? Yes / No

Languages

L is Recursive

halts on all inputs / in accept state $w \in L$

& halts in non-accept state if $w \notin L$

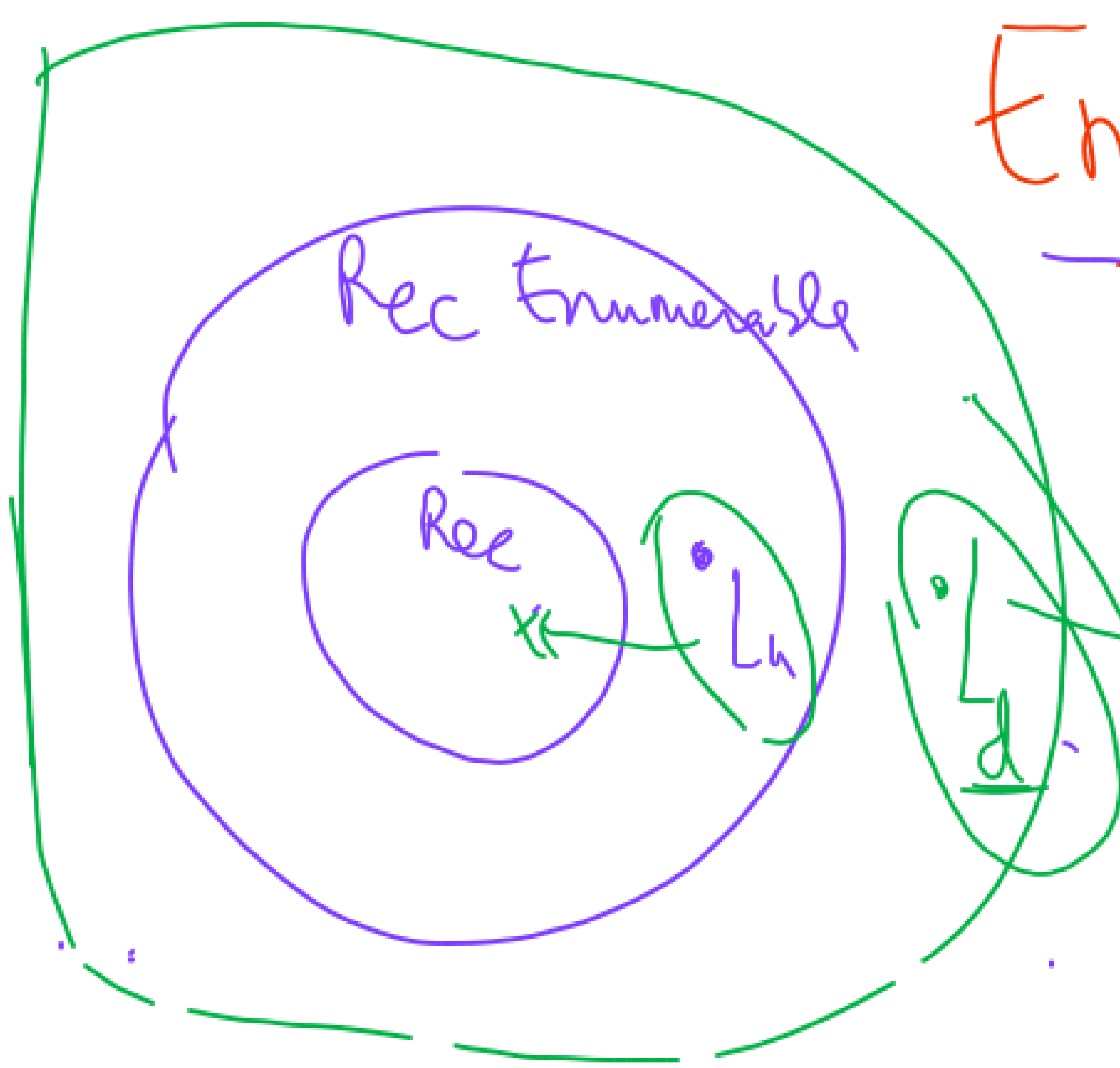
||

L is Recursively Enumerable

if there is a TM M such that M accepts L

Semi-decidable

Undecidable



"no TM can be built"

prime?

Programming

$\exists x^3y^2 + \dots \stackrel{?}{=} 0$

Hilbert's 10th prob